

UNIT V

Energy Sources and E-Waste Management

Define a fuel cell. Explain how a fuel cell differs from a conventional battery. Write the advantages and disadvantages of Fuel cells.

A fuel cell is a galvanic cell in which the chemical energy in an oxidant system is directly converted into electrical energy through the electrochemical process in which fuel is oxidised at the anode and oxidant is reduced at the cathode.

Like other electrochemical cells, the fuel cell has two electrodes and an electrolyte. The general representation of a fuel cell is-



General electrode reactions:

At anode: fuel undergoes oxidation, $\text{fuel} \rightarrow \text{oxidized product} + n\text{e}^{-1}$

At anode: the oxidant gets reduced $\text{oxidant} + n\text{e}^{-1} \rightarrow \text{reduced products}$

The electrons liberated from the oxidation process at the anode can perform useful work when they pass through the external circuit to the cathode.

Difference between a fuel cell and a conventional battery:

Fuel Cell	Battery
1. Fuel cell produces electrical energy continuously as long as the fuel and the oxidants are fed into the respective electrodes.	1. The battery produces electrical energy until the materials are present in the cell
2. Reactants are fed from outside to the electrode of the fuel cell	2. Reactants are an integral part of the construction of the cell
3. Fuel cell do not require recharging	3. Battery needs recharging.
4. High efficiency of production of electrical energy	4. Low efficiency of production of electrical energy.
5. Fuel cells need less attention.	5. Battery needs more attention.
6. Waste production in the fuel cell are harmless	6. Waste products in the battery may be harmful
7. Fuel cells do not store chemical energy.	7. Batteries store chemical energy.

Advantages of fuel cells:

1. They produce continuous energy.
2. Power efficiency is high
3. They are eco-friendly as the products of the overall reactions are not toxic, and hence no environmental problems.
4. Silent operations
5. It produces harmless waste products
6. No need to charge

Explain the construction, operation, reactions and applications of the methanol-oxygen fuel cell.

Methanol is one of the most electroactive organic fuels in the low-temperature range because 1. It has a low carbon content. 2. It possesses a readily oxidizable OH group. 3. It has high solubility in aqueous solutions.

In this, the anode consists of a porous nickel sheet on which platinum–palladium catalyst is deposited, and the cathode consists of a silver-impregnated porous nickel sheet. A membrane is inserted adjacent to the cathode on the electrode side. To minimize diffusion and, as a result, reduce the presence of methanol near the cathode. The electrolyte containing methanol is circulated through the anode chamber. Pure oxygen or air is passed through the cathode chamber, and sulphuric acid, which acts as the electrolyte, is placed in the central compartment.

At anode: $\text{CH}_3\text{OH} + \text{H}_2\text{O} \rightarrow 6\text{H}^+ + \text{CO}_2 + 6\text{e}^-$

Water is consumed at the anode in the reaction, and Hydrogen gas is produced. Similarly, oxygen enters the cathode electrode and forms water.

At cathode: $3/2 \text{O}_2 + 6\text{H}^+ + 6\text{e}^- \rightarrow 3\text{H}_2\text{O}$

Overall reaction: $\text{CH}_3\text{OH} + 3/2 \text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{CO}_2$

Uses: It is used in military applications and large-scale power production

Disadvantages:

- (i) **The cost** of power is high as a result of the cost of electrodes
- (ii) Fuel and oxidants need to be stored in tanks under high pressure. To have an appropriate voltage, battery fuel cells must be available

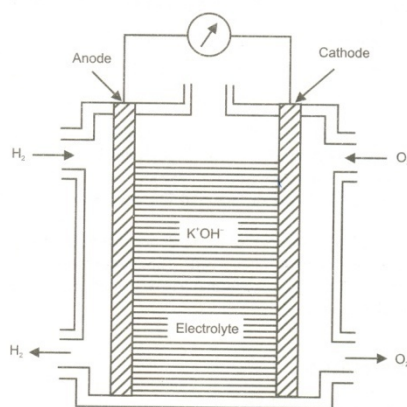


Figure: Methanol-oxygen fuel cell.

Justify the statement that Fuel cells are only energy conversion devices and not energy storage devices.

In fuel cells, reactants (fuel and oxidant) are fed into the cell. They do not store chemical energy and are capable of supplying current as long as they are supplied with the reactants.

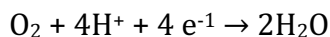
Polymer electrolyte membrane (PEM) fuel cells: This ion exchange membrane is used as a solid electrolyte for ionic conduction between anode and cathode, and also functions as a separator for the two reactant gases. The electrodes are made of a platinum block, and noble metals are used as electrocatalysts.

Electrode reactions:

Hydrogen-oxygen PEM fuel cell. Hydrogen molecules adsorb at the anode and are oxidized to protons. Electrons travel through an external load resistance. Protons diffuse through the PEM under an electrochemical gradient to the cathode. Oxygen molecules adsorb at the cathode, are reduced and react with the protons to produce water. The product water is absorbed into the PEM or evaporates into the gas streams at the anode and cathode.

At anode: catalytic oxidation of hydrogen $2\text{H}_2 \rightarrow 4\text{H}^+ + 4\text{e}^-$

At cathode: combining catalytically with oxygen and electrons to form water.



- Advantages:**
1. No free corrosive liquid in the cells
 2. Simple to fabricate
 3. Able to withstand large pressure
 4. Long life.

- Disadvantages:**
1. Electrolytes are expensive, and cell cost is high
 2. CO tolerance is poor.

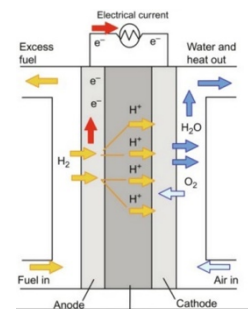


Figure: Polymer electrolyte membrane (PEM) fuel cells.

Solar energy

Introduction: Solar energy is radiant light and heat from the Sun that is harnessed using a range of technologies such as solar power to generate electricity, solar thermal energy (including solar water heating), and solar architecture. It is an essential source of renewable energy, and its technologies are broadly characterized as either passive solar or active solar depending on how they capture and distribute solar energy or convert it into solar power. Active solar techniques include the use of photovoltaic systems, concentrated solar power, and solar water heating to harness the energy.

A solar cell or photovoltaic cell is an electronic device that converts sunlight directly into electricity by the photovoltaic effect, which is a physical and chemical phenomenon.

When light shines on a photovoltaic (PV) cell that light may be reflected, absorbed, or passes right through the cell. The PV cell is composed of semiconductor material; the “semi” means that it can conduct electricity better than an insulator but not as well as a good conductor like a metal. There are several different semiconductor materials used in PV cells.

Importance of solar PV cells

Photovoltaic cells can generate electricity for a wide variety of applications:

1. They can be used in domestic lighting, spinning of fans, transistor radios, small TV sets, and tape recorders.
2. TV, internet, loudspeakers, VCR, and satellite linking for education; in health centres for surgery, for refrigerators and sterilizers; for mass communication, in telephones, broadcasting stations and in street lighting.
3. In agricultural sectors, PV cells can be used for irrigation, and in animal husbandry for watering and milking.
4. Electricity can be generated in rural areas, for individual families who are living far away from an electric grid connection and in remote areas such as mountains.
5. In production activities such as milling, sawing and sewing.

Construction and working of a solar PV cell

A photovoltaic cell mainly consists of a p-n junction diode made from Si semiconductors. It has two electrical contacts; one is a metallic grid over the n-type and the second is a layer of silver metal at the back of the p-type semiconductor. An anti-reflective layer (Silicon nitride or TiO_2) is coated between the metal grids to prevent the reflection of solar light.

When sunlight with its photons carrying energy $E = h\nu$ (which is enough to cross the barrier potential), falls on the p-n diode, electrons are ejected from the p-type semiconductor, creating an electron-hole pair. These ejected electrons move to the n-diode through the p-n junction. Since the p-n junction allows one-way movement of electrons, these electrons must flow through the external circuit to recombine with holes present in the p-side of the semiconductor. This movement of electrons from the n-side to the p-side through the external circuit generates an electric current.

Electrical energy obtained from a PV cell is directly proportional to the area of the cell exposed to light radiation and the intensity of light radiation, which can eject electrons.

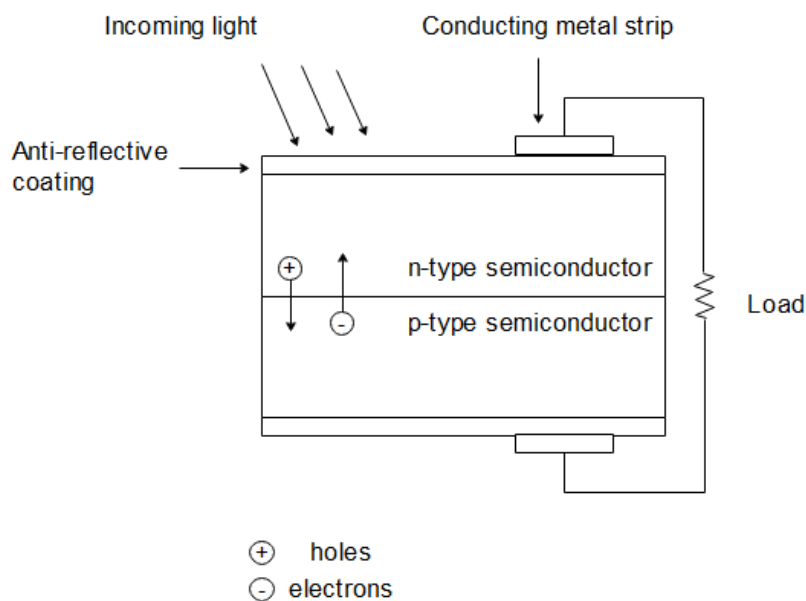


Figure: Photovoltaic cell

Advantages of a solar PV cell

1. Does not require any recharging.
2. They do not corrode.

3. They are environmentally friendly.
4. Make use of renewable sources of energy.
5. Low operating cost (no fuel).
6. They operate at ambient temperature (no high-temperature corrosion or safety issues).
7. They do not suffer wear and tear as they have no movable parts.
8. It is a non-polluting, low-maintenance energy source.
9. Quick installation.
10. High public acceptance and excellent safety record.

Disadvantages of solar PV cells

1. Works only in the presence of sunlight.
2. High installation costs.
3. The power efficiency of a commercial Si-based PV cell is very low (16%).
4. The silicon required for a PV cell is of high purity (99.999%). The high cost of Si purification has resulted in higher prices of crystalline silicon solar cells.
5. Energy can be produced only during the daytime.
6. Lack of widespread commercially available system integration and installation.
7. Space is a constraint, that is space required to generate unit power output is relatively more.
8. Dust often accumulates on the panel, thus reducing its efficiency.

Hydrogen Energy

Introduction:

Hydrogen is emerging as a promising energy source due to its potential to provide clean, efficient, and sustainable energy. As the most abundant element in the universe, hydrogen can be utilized in various forms to produce power, heat, and fuel, making it a key player in the transition to a low-carbon economy.

Production of Hydrogen

Hydrogen can be produced using several methods:

1. **Electrolysis:** Water (H₂O) is split into hydrogen and oxygen using electricity. When powered by renewable sources like wind or solar, this method produces "green hydrogen," a clean and sustainable form of hydrogen.
2. **Steam Methane Reforming (SMR):** This process extracts hydrogen from natural gas but produces carbon dioxide (CO₂) as a byproduct. This method is currently the most common but is less environmentally friendly due to CO₂ emissions.
3. **Coal Gasification and Biomass Conversion:** Other methods involve breaking down coal or biomass to produce hydrogen, although they are typically less efficient and environmentally friendly compared to electrolysis.

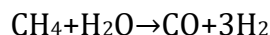
I. Steam Methane Reforming (SMR) for Hydrogen Production

Steam Methane Reforming (SMR) is the most widely used industrial process for producing hydrogen from natural gas. It involves the reaction of methane (CH₄) with steam (H₂O) over a catalyst at high temperatures to produce hydrogen gas (H₂) and carbon monoxide (CO). This process is key in industries such as ammonia production, petroleum refining, and energy generation.

Procedure

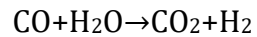
SMR is a two-step process that typically occurs in two reactors:

1. **Reforming Reaction (Primary Stage):**
 - Methane reacts with steam at high temperatures (700-1000°C) in the presence of a catalyst, typically nickel-based.
 - The reaction produces hydrogen (H₂) and carbon monoxide (CO):



2. **Water-Gas Shift Reaction (Secondary Stage):**
 - The carbon monoxide produced in the first reaction reacts with more steam at a lower temperature (200-450°C) in the presence of a catalyst (typically

iron oxide or chromium oxide) to produce additional hydrogen and carbon dioxide (CO₂):



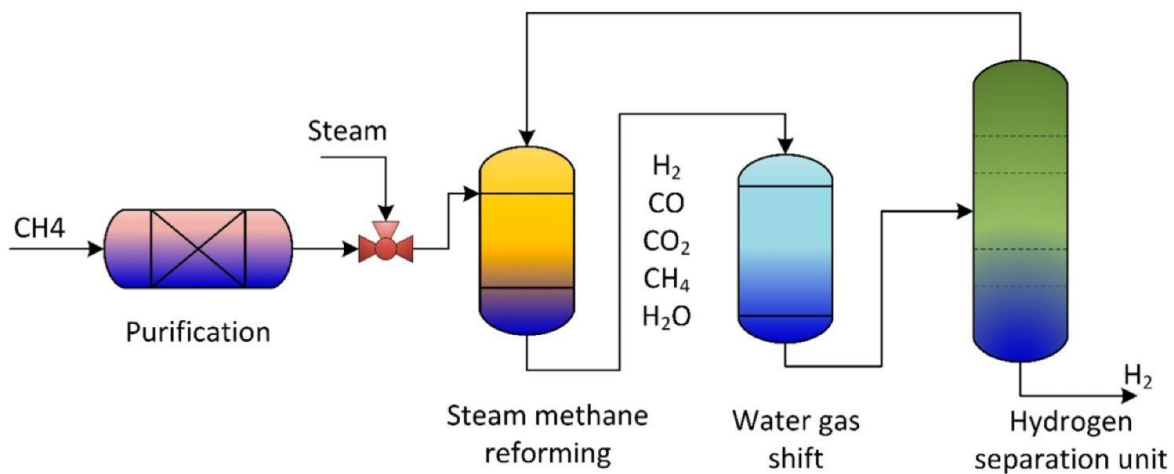
Reaction Conditions

- Temperature: 700-1000°C
- Pressure: 3-25 bar (high pressure is typically used to enhance hydrogen production)
- Catalyst: Nickel or a nickel-based catalyst for the reforming reaction; iron or copper-based for the shift reaction.

Purification of H₂

After the reforming and shift reactions, the gas mixture contains hydrogen, carbon dioxide, carbon monoxide, and other impurities. Hydrogen purification is necessary to isolate high-purity hydrogen, typically achieved through processes like:

- Pressure Swing Adsorption (PSA): Separates hydrogen from other gases based on their adsorption properties.
- Membrane Separation: Uses selective permeability to separate hydrogen from other gases.



II. Hydrogen production using electrochemical and photoelectrochemical Methods

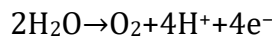
Hydrogen production using electrochemical and photoelectrochemical water splitting are two promising methods for generating clean hydrogen from water using renewable energy sources. Both processes involve splitting water (H_2O) into hydrogen (H_2) and oxygen (O_2), but they differ in the energy sources used to drive the reactions.

1. Electrochemical Water Splitting

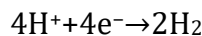
In electrochemical water splitting, an electric current is applied to water to break it into hydrogen and oxygen. This process typically takes place in an electrolyzer, which consists of two electrodes: the anode (where oxygen is produced) and the cathode (where hydrogen is produced).

Steps in Electrochemical Water Splitting:

- Anode reaction (oxygen evolution reaction - OER): At the anode, water is oxidized to produce oxygen gas, protons, and electrons:



- Cathode reaction (hydrogen evolution reaction - HER): At the cathode, protons from the electrolyte combine with electrons to form hydrogen gas:

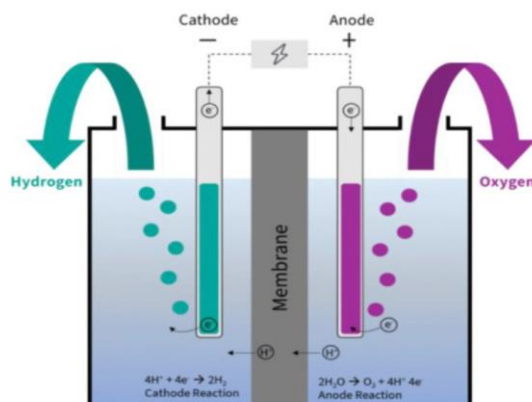


Energy Source:

- The electricity required for electrolysis can be derived from renewable sources such as wind, solar, or hydropower. This ensures that the process can be sustainable and low in carbon emissions.

Electrolyzer Types:

Proton Exchange Membrane (PEM) Electrolyzers: Use a solid polymer membrane as the electrolyte, offering higher efficiency and faster response times.

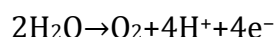


2. Photoelectrochemical Water Splitting (PEC):

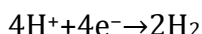
Photoelectrochemical water splitting uses solar energy to directly drive the splitting of water into hydrogen and oxygen. The process combines photovoltaic (PV) technology and electrochemical water splitting in a single device, called a photoelectrochemical cell (PEC).

Steps in Photoelectrochemical Water Splitting:

- Light absorption: A semiconductor material in the PEC absorbs sunlight, generating electron-hole pairs (excited electrons and holes).
- Charge separation and transfer: The excited electrons and holes are separated and directed to the anode and cathode, respectively.
- Anode reaction (Oxygen Evolution Reaction - OER): At the anode, the electrons from the semiconductor are used to oxidize water, producing oxygen:



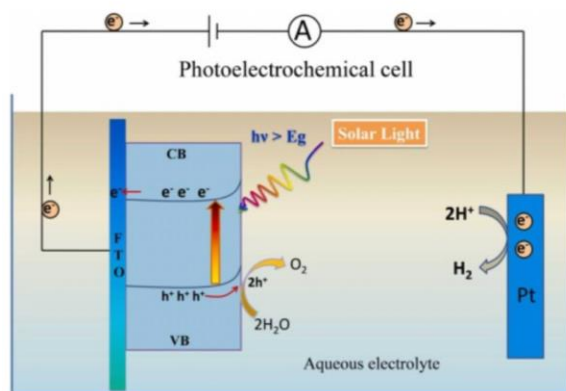
- Cathode reaction (Hydrogen Evolution Reaction - HER): At the cathode, the holes combine with electrons from the external circuit to reduce protons (from the water) to form hydrogen gas:



Materials Used in PEC:

- Semiconductors: Materials like silicon (Si), titanium dioxide (TiO_2), and others are used to absorb light and facilitate the separation of charge carriers. The efficiency of the photoelectrode depends on its ability to absorb light across the solar spectrum and its stability in aqueous environments.
- Electrocatalysts: Catalysts are needed at both the anode and cathode to enhance the reaction rates. These include materials like platinum (Pt), iridium oxide (IrO_2), and nickel-based compounds.

Energy Source: Solar energy directly drives the reaction. The efficiency of PEC systems is highly dependent on the material properties of the photoelectrode and the overall cell configuration.



Comparison of Electrochemical vs. Photoelectrochemical Water Splitting

<i>Aspect</i>	<i>Electrochemical Water Splitting</i>	<i>Photoelectrochemical Water Splitting</i>
Energy Source	Electricity (from renewable sources)	Solar energy (directly drives the reaction)
Efficiency	High efficiency with optimized electrolyzers	Lower efficiency due to energy losses in light absorption and charge transfer
Technology Readiness	Well-developed with commercial systems	Still in research and development stages
Materials Required	Electrodes, electrolytes, catalysts	Semiconductor materials, catalysts
System Complexity	Simple, mature technology	More complex, requires advanced materials
Scalability	Easily scalable for large-scale hydrogen production	Less scalable due to material and efficiency challenges

Challenges:

- **Electrochemical Water Splitting:** Challenges include the need for efficient, stable, and cost-effective catalysts for both the oxygen and hydrogen evolution reactions, as well as the energy input required for electrolysis.
- **Photoelectrochemical Water Splitting:** The main challenges are low efficiency, stability of photoelectrode materials under long-term solar exposure, and the complexity of integrating efficient catalysts with semiconductor materials.

Hydrogen Storage:

Hydrogen storage is a critical challenge in the development of hydrogen fuel technologies, especially for mobile applications such as fuel cell vehicles. Several materials are being explored for their ability to store hydrogen efficiently, with different types offering distinct advantages and challenges. These materials generally fall into three categories: metal hydrides, high surface area materials, and complex and chemical hydrides.

1. *Metal Hydrides*

Metal hydrides are compounds formed by metals and hydrogen. These materials absorb hydrogen gas at relatively low pressures and release it upon heating or under the application of a catalyst. Metal hydrides are one of the most studied hydrogen storage options due to their relatively high volumetric energy density and reversible hydrogen absorption.

- **Example Materials:**

- Magnesium hydride (MgH_2): Known for its high hydrogen storage capacity by weight (around 7.6% by weight) and a relatively low desorption temperature, magnesium hydride has been extensively researched.
- Sodium alanate (NaAlH_4): This material can release hydrogen at moderate temperatures (around 200-300°C) and is of interest for both mobile and stationary applications.
- Lanthanum nickel hydride (LaNi_5H_6): Another well-known material with good hydrogen storage characteristics, although it suffers from high cost and limited hydrogen capacity.
- Advantages:
 - High volumetric density of hydrogen.
 - Reversible hydrogen absorption and desorption processes.
 - Relatively safe to handle compared to high-pressure gaseous storage.
- Challenges:
 - High desorption temperatures (especially for magnesium hydride).
 - Slow kinetics for hydrogen absorption and desorption.
 - Need for thermal management systems.

2. *High Surface Area Materials*

Materials with high surface area are typically porous materials like activated carbon, metal-organic frameworks (MOFs), and zeolites. These materials store hydrogen through physisorption (weak van der Waals forces) rather than chemical bonding, which allows for easier reversibility.

- Example Materials:
 - Activated Carbon: The high surface area of activated carbon allows it to adsorb hydrogen gas at relatively low pressures and ambient temperatures.
 - Metal-Organic Frameworks (MOFs): MOFs are a class of crystalline compounds composed of metal ions and organic ligands, which have enormous surface areas and tunable structures. Some MOFs are capable of adsorbing large amounts of hydrogen at moderate pressures and low temperatures.
 - Zeolites: These crystalline, microporous materials also show potential for hydrogen storage due to their high surface area and tunable pore sizes.
- Advantages:
 - Lightweight and can be tailored to enhance hydrogen adsorption capacity.

- Operate at lower temperatures and pressures compared to chemical hydrides.
- Fast kinetics for hydrogen absorption and desorption.
- Challenges:
 - Hydrogen density is lower compared to metal hydrides.
 - Adsorbed hydrogen is less stable and may desorb at lower temperatures or pressures.
 - Still requires relatively high pressures for significant storage capacity.

3. *Complex and Chemical Hydrides*

Complex hydrides are materials that combine metal hydrides with additional elements, forming compounds that can absorb and release hydrogen more efficiently at lower temperatures. Chemical hydrides, such as ammonia borane (NH_3BH_3), involve chemical reactions that release hydrogen when activated by heat or a catalyst.

- Example Materials:
 - Ammonia Borane (NH_3BH_3): This chemical hydride releases hydrogen through a process called hydrolysis or dehydrogenation and can release up to 19.6% hydrogen by weight. However, handling and regeneration remain challenges.
 - Sodium borohydride (NaBH_4): Sodium borohydride is another chemical hydride that stores hydrogen through a similar dehydrogenation process. It has high hydrogen content but requires high activation energy for hydrogen release.
 - $\text{LiNH}_2\text{-LiH}$: Lithium amides and hydrides have been combined to form complex hydrides capable of storing hydrogen at moderate temperatures.
- Advantages:
 - High hydrogen density by weight, especially in chemical hydrides like ammonia borane.
 - Can operate at lower temperatures compared to traditional metal hydrides.
 - More energy-efficient in certain configurations.
- Challenges:
 - Requires specific conditions (temperature, pressure, or catalyst) for hydrogen release.
 - Often difficult to regenerate or recover hydrogen efficiently.

- Safety concerns due to the chemical nature of some hydrides.

Advantages of Hydrogen

- Clean Energy: Hydrogen combustion or fuel cell use produces only water vapor as a byproduct, making it a zero-emission energy source.
- High Energy Density: Hydrogen has a high energy content per unit of weight, making it ideal for long-distance transportation and heavy-duty applications.
- Versatility: Hydrogen can be used in various sectors, from transportation to industrial processes, and can complement renewable energy sources.

Challenges of Hydrogen as an Energy Source

- Production Cost: Green hydrogen produced via electrolysis is currently more expensive than hydrogen derived from natural gas (gray hydrogen), limiting its widespread adoption.
- Infrastructure Needs: Developing the infrastructure for hydrogen production, storage, and distribution is costly and complex.
- Storage and Transport: Hydrogen is a low-density gas, so it requires advanced storage solutions or conversion into liquid form, posing logistical challenges.